Towards Practical Intelligent CPU Cooling: Simulation-Driven Adaptive Fan Control with State-Machine Regularized Online RL

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***Abstract*—Intelligent CPU cooling adaptively adjusts fan speeds to keep temperatures within safe limits while minimizing fan power. However, realizing intelligent CPU cooling in production systems faces several practical challenges: prohibitively long iteration cycles for algorithm deployment, the need for cooling-specific controllers under bounded PWM actuation, and time-varying cooling dynamics that demand safe adaptation after deployment. We propose an end-to-end framework for practical intelligent CPU cooling. Firstly, we build a high-fidelity simulation environment with learned temperature and fan-power predictors to enable rapid policy iteration. We then propose Prior-Regularized Policy Optimization (PRPO), a cooling-specific RL algorithm for bounded fan control that parameterizes actions in the physical range and regularizes updates toward rule-based priors to stabilize learning and reduce oscillatory exploration. Finally, we introduce a state-machine-regularized online adaptation scheme that constrains when and how the controller updates, enabling safe adaptation to drift after deployment. Experiments show that our simulator achieves a temperature prediction MAE of as low as 1.83°C and fan-power prediction R^2 exceeding 0.9. Moreover, in both simulated and real environments, our controller consistently outperforms state-of-the-art controllers, such as BP-PID and I-PID, achieving the lowest comprehensive control cost.**

I. Introduction

With the rapid growth of cloud and high-performance computing, server CPUs face increasingly intensive workloads, causing heat accumulation that degrades performance and reliability [1], [2]. **Intelligent CPU cooling** addresses this issue by adjusting fan speeds according to real-time CPU states. As shown in Fig. 1, the control algorithm runs on the baseboard management controller (BMC) [3], which monitors temperature and fan power and generates PWM signals to maintain CPU temperature below a preset threshold while reducing fan power.

Practical deployment remains challenging because: 1) BMC implementation often requires manual transcription, resulting in long iteration cycles; 2) bounded PWM control and smoothness constraints are poorly handled by off-the-shelf RL [4]–[9], especially because unconstrained exploration can produce unsafe or oscillatory actions [10], [11]; and 3) cooling dynamics vary over time due to hardware aging and environmental changes.

To address these challenges, we propose an end-to-end framework consisting of: 1) a simulation environment with a Temporal Fusion Network (TFN) for temperature prediction and a lightweight LSTM for fan-power prediction; 2) Prior-Regularized Policy Optimization (PRPO) for bounded PWM control; and 3) finite-state-machine-regularized online adaptation. The simulator achieves a temperature-prediction MAE of 1.83°C and fan-power prediction $R^2 > 0.9$. The proposed controller also achieves the best overall performance in both simulated and real environments compared with Naive-PID [12], BP-PID [13], and I-PID [14].

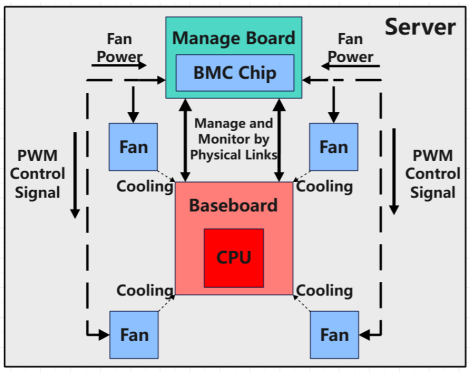


Fig. 1. Typical CPU cooling system. Solid lines denote monitoring signals, while dashed lines denote PWM control outputs.

II. Problem Formulation

The CPU cooling problem can be ideally formulated and depicted using the following time series:

$$(W_1, P_1, T_1, E_1),\ (W_2, P_2, T_2, E_2),\ \ldots,\ (W_k, P_k, T_k, E_k) \tag{1}$$

where W_k represents the working parameters of the CPUs, such as CPU power, which indicates the current workload condition. P_k denotes the PWM wave values at time step k, serving as control signals. T_k indicates the resultant CPU temperature at time step k, and E_k represents the fan energy consumption at time step k. Generally,

$$T_k = F\left(\{W_t, P_t, T_t\}_{t=1}^{k-1}, W_k, P_k\right) \tag{2}$$

and the power consumption at time k is primarily relevant to the historical PWM values:

$$E_k = G\left(\{P_t\}_{t=1}^{k}\right) \tag{3}$$

In Eq. 2 and 3, F and G are unknown mappings.

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Our objective is to determine per-time-step PWM values, which govern the fan speed, to maintain CPU temperature below the preset threshold T_{set} (i.e., the setpoint) while minimizing fan power consumption. This can be expressed mathematically as:

$$\begin{aligned} \min_{\{P_t\}_{t=1}^{k}} \quad & \overline{M} = \frac{1}{k}\sum_{t=1}^{k} E_t \\ \text{s.t.} \quad & T_t \le T_{\text{set}}, \quad \forall t = 1, 2, \ldots, k \end{aligned} \tag{4}$$

In Eq. 4, $\overline{M}$ denotes the average fan power dissipation. In practice, brief excursions above the setpoint may be tolerated due to sensor noise and transient workload changes; therefore, the setpoint is treated as the primary operating target rather than as a strict hard guarantee.

III. Simulation environment construction

TFN For Temperature Prediction. In our simulation environment, F is constructed by a temporal fusion network (TFN). The TFN is characterized by a two-branch architecture. The first branch employs a long short-term memory (LSTM) [15]–[18] module to encode multi-dimensional time series from the initial step to the $(k-1)$-th time step. At the current time step k, input values W_k and P_k are processed via multi-layer perceptron (MLP) [19] layers to generate a vector embedding. Features from the previous and current time series are fused using an attention module [20]–[23], followed by a linear layer for dimensionality reduction—ultimately outputting the predicted temperature T'_k.

This architecture effectively extracts and fuses temporal features (past and current) to enable high-fidelity temperature prediction. Notably, with an input time sequence length of 5, the TFN only contains 38K parameters and requires 0.11M floating-point operations (FLOPs), ensuring fast temperature prediction and frequent invocations during controller training.

Enhanced Loss For TFN Training. Practically, the collected time series described in Eq. 1 contain noise. Additionally, the trained TFN must adhere to a fundamental physical principle: under stable CPU workloads, increased PWM values probably correspond to lower CPU temperature, as a larger PWM value drives a faster fan speed. In light of this, physics-informed learning provides a principled way to incorporate physical knowledge into data-driven models [24]. We therefore propose the training loss $\mathcal{L}$ for our TFN as follows:

$$\mathcal{L} = \mathcal{L}_{\text{smoothL1}} + \mathcal{L}_{\text{physics}} \tag{5}$$

In Eq. 5, $\mathcal{L}_{\text{smoothL1}}$ denotes the smoothed L1 loss, which is generally considered more robust to noise-induced outliers in time series during training. Moreover, $\mathcal{L}_{\text{physics}}$ is defined as:

$$\mathcal{L}_{\text{physics}} = \alpha \cdot \max(0, \rho(P_k, T'_k)) \tag{6}$$

where ρ represents the Pearson correlation coefficient between the input PWM value P_k and the predicted temperature T'_k. This loss activates only when P_k and T'_k show a positive correlation, which contradicts the stated physical principle, thereby suppressing the non-physical behavior. The α serves to scale the magnitude of $\mathcal{L}_{\text{physics}}$.

Power Consumption Predictor. Since fan power consumption is solely a function of PWM values, as defined in Eq. 3, G is formed by using a single LSTM module with the PWM sequence as input and the predicted fan power consumption as output. This manner is simple yet effective, as validated in Experimental Evaluation.

IV. RL Controller Trained in Simulation

Policy-gradient and actor–critic methods, including PPO, DDPG, SAC, and TD3, are widely used for continuous-control tasks [9], [25]–[27]. PPO variants have also been adopted in diverse decision-making applications [28]–[31]. Our PRPO is composed of three mechanisms:

Topological Consistency via Beta Distribution. Conventional continuous control policies typically assume actions follow a Gaussian distribution. However, PWM control signals are physically confined to a compact interval of [0,1]. Gaussian policies necessitate hard clipping when handling saturation states (Full Speed 1.0 or Stop 0.0). This operation induces gradient vanishing during backpropagation, rendering the agent incapable of learning critical boundary policies. To address this, we reconstruct the policy network using the Beta distribution, which is topologically homeomorphic to the finite interval [0,1]. This formulation not only eliminates the gradient bias caused by clipping but also allows the model to precisely regulate the output confidence.

Asymmetric Reward. We use a composite reward

$$R_t = R_{temp} + R_{energy} + R_{smooth} \tag{7}$$

where R_{temp} penalizes temperature overshoot ($T > T_{set}$) much more strongly than undercooling, thereby prioritizing thermal safety while allowing energy-efficient operation near the setpoint. We further define $R_{energy} = -\lambda_e A_t$ to penalize high fan speed and $R_{smooth} = -\gamma|A_t - A_{t-1}|$ to discourage abrupt PWM changes. This reward encourages safe, energy-efficient, and smooth control.

Dynamic Setpoint Training. To ensure satisfactory controller performance across multiple setpoints, we assign different setpoints proportionally to various data segments in the training dataset. Simultaneously, the reward function is adjusted in tandem with setpoint changes to enable dynamic functionality.

V. Online learning of trained RL controller

Simulation-to-real transfer commonly introduces randomized or diversified simulated conditions to improve robustness under deployment mismatch [32]. Our online stage instead uses rule-guided adaptation and state-machine constraints to address time-varying cooling dynamics safely.

Safe Knowledge Internalization via Shadow Mode. To eliminate the "cold-start" instability inherent in on-policy learning, we employ a "Shadow Mode" mechanism. In this phase, the RL controller operates passively alongside the deterministic rule-based controller. We formulate this mechanism as a Maximum Likelihood Estimation (MLE) problem. The controller treats the output of the deterministic rules as ground-truth samples and updates its parameters to maximize the probability of reproducing these expert behaviors. This ensures that the controller's initial policy manifold aligns with stable baselines before active control is granted.

Dynamic Logical Regularization. The Finite State Machine (FSM) functions as a dynamic "Logical Regularizer"

that arbitrates control authority based on the real-time system status. Specifically, the FSM enforces deterministic logic during steady-state operations to maximize energy efficiency and minimize variance. Conversely, it activates the PRPO inference only during transient workload spikes where nonlinear adaptability is required. This mechanism effectively utilizes "hard" logic to regulate "soft" exploration, achieving an optimal balance between global stability and transient agility. The resulting action defined as:

$$u_{final}(t) = \mathbb{I}(s_t \in \mathcal{S}_{stable}) \cdot u_{rule}(s_t) + \mathbb{I}(s_t \in \mathcal{S}_{transient}) \cdot u_{\pi}(s_t|\theta) \quad (8)$$

where $u_{final}(t)$ represents the actual control command applied to the fans at time t; s_t is the current environmental state; $\mathcal{S}_{stable}$ and $\mathcal{S}_{transient}$ denote the state subspaces defined for steady and transient regimes, respectively; $\mathbb{I}(\cdot)$ is the indicator function; u_{rule} is the output from the deterministic controller; and u_{π} is the RL policy output.

VI. Experimental Evaluation

A. Dataset Collection and Augmentation

Real-World Data Collection. We collected a total of 7 hours of time-series data from the physical server cluster to train and validate our simulation models (S_1 and S_2). The data collection protocol was designed to cover a comprehensive state space. The collected dataset was split into a 70% training set for training the simulator parameters and a 30% validation set to verify prediction accuracy.

Simulation-Based Data Augmentation. Our simulator allows us to generate an unlimited amount of diverse interaction trajectories for RL training, exposing the controller to complex corner cases that may rarely occur in the limited 7-hour real-world dataset. This strategy can significantly enhances the generalization capability of the controller.

B. Evaluation Metrics

To evaluate the performance of our simulation environment, we employ two metrics. The first metric is the well-known mean absolute error (MAE). The second metric is the maximal temperature difference (MTD), defined as:

$$\text{MTD} = \max_{t=1,2,\ldots,k} \left(T_t^1 - T_t^{100}\right) \quad (9)$$

In Eq. 9, T_t^1 represents the temperature sequence when the controller's PWM values are consistently set to 1, while T_t^{100} denotes the temperature sequence when the PWM values are consistently set to 100.

To evaluate controllers, we adopt three physical metrics. The first two are defined as:

$$M_1 = \frac{1}{k}\sum_{t=1}^{k} |T_t - T_{\text{set}}| \quad (10)$$

$$M_2 = \max_{t=1,\ldots,k} (T_t - T_{\text{set}}) \quad (11)$$

Here, M_1 measures the average absolute tracking error, M_2 identifies the most extreme signed deviation of temperature from the setpoint during the observation period. The third metric is the average fan power consumption $\overline{M}$, as defined in Eq. (4). Notably, M_1 and M_2 are quantified in degrees Celsius (°C), while $\overline{M}$ is measured in watts (W).

Last but not least, we define Comprehensive Control Cost (3C) to facilitate a uniform comparison of different controllers. The 3C is formulated as:

$$3C = \frac{1}{3}\text{Norm}(M_1) + \frac{1}{3}\text{Norm}(M_2) + \frac{1}{3}\text{Norm}(\bar{M}) \quad (12)$$

Notably, the term Norm refers to the Min-Max normalization [33] operation among all the controllers for comparison. Besides, we use equal weighting to avoid introducing a post hoc preference for any single criterion. **Importantly, 3C does not replace the raw physical metrics; rather, it complements them by revealing which controller achieves the best overall thermal-safety–energy trade-off.**

Below, all metric results based on the collected data are averaged over multiple independent runs. Accordingly, the differences among them are also stable.

C. Results on Simulation Environment

Advantages of TFN's Two-Branch Structure. Table I clearly demonstrates that none of the double branches in our TFN is dispensable. Whether only the MLP branch is retained for extracting current features or only the LSTM branch for capturing past time features, the predicted MAE of the TFN significantly increases. This finding indicates that the predicted temperature T_k' at time step k is influenced by both current and past time features. Furthermore, we investigate the correlation between the predictions of the TFN and the number of past time steps, denoted as ts in Table I. The results show that a value of $ts = 5$ yields the most accurate temperature predictions.

TABLE I
Results on TFN Validity

Model Architecture	Mean Absolute Error
Only MLP Branch	5.67
Only LSTM Branch	4.77
TFN (Time Step = 1)	2.71
TFN (Time Step = 2)	2.79
TFN (Time Step = 5)	**1.83**
TFN (Time Step = 10)	1.92

Effect of Enhanced Loss. The MTD is evaluated in three scenarios: 1) As-is scenario: In this scenario, the input data is entirely sourced from the validation set, where the CPU power is not constant. 2) Fixed CPU power scenario: Here, the current CPU power W_k is held constant, using the average CPU power derived from the validation set. 3) Synthetic scenario: In this case, all values W_t for $t = 1$ to k are set to the average CPU power from the validation set, making it temporally constant. The rationale behind these scenarios is that a more constant CPU power should reinforce the negative correlation between the input PWM values and predicted temperature. As shown in Table II, without our proposed enhanced loss $\mathcal{L}_{\text{physics}}$, the MTD is negative in all three scenarios, indicating a violation of physical principles. This underscores our enhanced loss function ensures the physical trustworthiness of F.

TABLE II
MTD ($^\circ C$) under different test data scenarios

Test Dataset Scenario	As-is	Fixed CPU Power	Synthetic
Without $\mathcal{L}_{\text{physics}}$	-0.46	-0.22	-1.19
With $\mathcal{L}_{\text{physics}}$	**6.68**	**6.89**	**5.38**

Advantages of Power Consumption Predictor. Fig. 2

demonstrates that our power consumption predictor G exhibits a significantly higher R^2 value (greater than 0.9) compared to the square predictor, which is based on the general assumption that PWM and power have a square relationship.

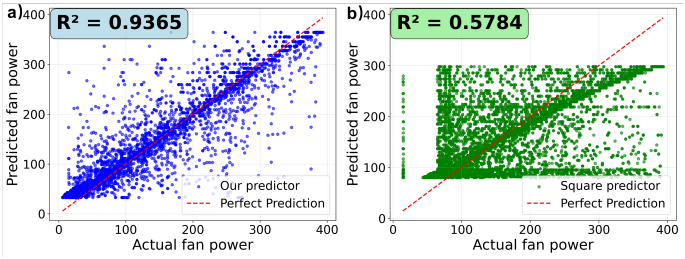


Fig. 2. Comparison of power prediction accuracy: a) Our predictor. b) Square predictor.

D. Controller's Performance in ***Real*** Environment

As shown in Table III, in **real** enviroments, our RL controller demonstrated the superior performance, achieving the lowest values in metrics $3C$, M_1 and $\bar{M}$. Our controller is capable of regulating the temperature to be around the setpoint with the minimum power consumption.

TABLE III
COMPARISON OF CONTROLLERS IN REAL ENVIRONMENTS

Controller	$M_1 \downarrow$	$M_2 \downarrow$	$\bar{M} \downarrow$	$3C \downarrow$
Naive-PID	5.94	-3.0	15.73	0.67
BP-PID	1.74	1.0	14.56	0.30
DQN [34]–[37]	3.52	7.27	14.35	0.58
Ours	**0.90**	**1.45**	**13.95**	**0.14**

E. Controller's Performance in Simulation Environment

Thorough Comparison with Related Work. As listed in Table IV, the baselines include: Naive PID, which employs a simple incremental structure; BP-PID, which utilizes back-propagation neural networks to update parameters exclusively when the temperature exceeds the setpoint; and I-PID, which executes parameter updates at every time step. Deep Q-learning and its variants have also been widely studied in control and decision-making tasks [38]–[41]. Experimental results demonstrate that our controller undoubtedly achieves the optimal $3C$ among all evaluated approaches. Besides, as for multi-setpoint results shown in Table V, clearly our controller demonstrated superior performance over the I-PID controller in all cases.

TABLE IV
COMPARISON OF OUR RL CONTROLLER AND RELATED WORKS

Controller	$M_1 \downarrow$	$M_2 \downarrow$	$\bar{M} \downarrow$	$3C \downarrow$
Naive-PID	5.91	10.25	26.22	0.88
BP-PID	5.37	8.79	28.51	0.91
I-PID	1.24	7.0	22.14	0.23
Ours	**1.03**	**1.05**	**23.63**	**0.08**

Visualization. In Fig. 3, we present the PWM value curves generated by our RL controller and the state-of-the-art I-PID controller, along with the corresponding temperature curves, based on a segment of data. The I-PID controller first adjusts its parameters and then outputs PWM values indirectly through these parameters. This control mechanism results in slow and insensitive PWM changes, which lead to temperature overshoot, as demonstrated in Fig. 3(c).

In contrast, our controller directly outputs PWM values. This control approach offers the advantage of rapid PWM response with right-angle-like changes (Fig. 3(d)), enabling more timely temperature control and thus avoiding the overshoot phenomenon.

TABLE V
PERFORMANCE COMPARISON OF CONTROLLERS FOR MULTIPLE SETPOINTS IN SIMULATION

Setpoint	Controller	$M_1 \downarrow$	$M_2 \downarrow$	$\bar{M} \downarrow$	$3C \downarrow$
50	I-PID	1.70	13.0	42.39	0.61
	Ours	**4.20**	**5.58**	**33.93**	**0.33**
55	I-PID	1.46	10.0	30.53	0.33
	Ours	**0.44**	**2.95**	**32.52**	**0.15**
60	I-PID	1.24	7.0	22.14	0.23
	Ours	**1.03**	**1.05**	**23.63**	**0.08**
65	I-PID	1.10	3.0	16.43	0.41
	Ours	**1.08**	**1.06**	**16.93**	**0.13**
70	I-PID	1.0	0.84	12.15	0.35
	Ours	**1.44**	**0.55**	**12.01**	**0.23**

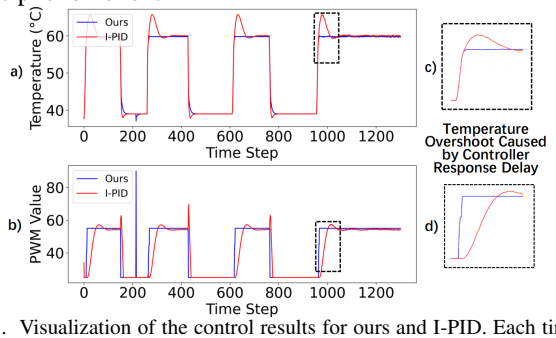


Fig. 3. Visualization of the control results for ours and I-PID. Each time step represents 1 second. a) Temperature response over time (Ours vs I-PID). b) PWM control signal over time (Ours vs I-PID). c) Zoom-in of a). d) Zoom-in of b).

Ablation Study on State Machine Strategy. Table VI shows that adding the state machine substantially improves control performance: M_1 decreases by 74.5% (from 4.84 to 1.24), M_2 drops from 10 to 2, and $\bar{M}$ is slightly reduced from 19.72 to 19.18. The resulting lowest $3C$ further supports the role of the state machine in stable and efficient thermal management. Note that a $3C$ score of 0 does not imply zero physical cost; it indicates the lowest relative cost after Min–Max normalization.

TABLE VI
ABLATION STUDY RESULTS ON STATE MACHINE STRATEGY

Method	$M_1 \downarrow$	$M_2 \downarrow$	$\bar{M} \downarrow$	$3C \downarrow$
PRPO_online with state-machine	**1.24**	**2**	**19.18**	**0**
PRPO_online w/o state-machine	4.84	10	19.72	1

VII. CONCLUSION

This work presents an end-to-end framework for intelligent CPU cooling. Our framework covers an efficient simulation environment, PRPO for bounded PWM fan control, and state-machine-regularized online adaptation. Experiments validate the simulator accuracy and demonstrate that the proposed controller consistently outperforms the state of art in both simulated and **real** environments, achieving a better trade-off between temperature regulation and fan energy.

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